Muhammad Alfarafi Maulana Firdausa

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Summary —

As a Core Network Engineer, I professionally design, implementation, maintenance, and troubleshooting of XL Axiata and TELKOM Indonesia core network infrastructure. I mostly work on project-based scenarios and small teams compared with the other accounts which require me to be as adaptable, efficient, fast, and accurate as possible. I have the key responsibilities of developing project solutions to meet Customer needs, planning and executing infrastructure expansion or new site solutions, Installing and Configuring network devices such as routers, switches, servers, and Storage, and Comisioning Virtual Network Functions such as MDAF(AI Network analyzer), FusionSphere(OpenStack), FusionStage(k8s), eSight, LCM, and OSMU, SBC, CSCF, ATS, and vDRA. I ensure the VoLTE and VoBB services have optimal network performance by monitoring and analyzing traffic, resolving issues, and maintaining service KPIs.

Skills

Core Network VoLTE, VoBB, IMS(P-CSCF, I-SCSF, S-CSCF, ATS, MRFP, CCF, DNS, Enum), SBC(A-SBC, I-SBC), SPS and USC(vDRA), EPC(CP/UP), NMS, MDAF, New Calling.

NFV VNF management, Hardware management, FusionSphere OpenStack, FusionStage, Virtualization Infrastructure of Network Functions, Kubernetes, Container runtime, Microservices Architecture.

Datacom L3 Router, L2 Lanswitch, Non-SDN, MPLS, OSPF.

Tools Dev Web development (Django, Python javascript), Mobile development (Kotlin, Java), IoT (Drone development, Smart Home).

Experience

Huawei Tech Investment **2022 – 2026**
Cloud Core Engineer (IMS VoLTE)

Business scenarios: Project Delivery

- Make data center delivery solutions, this includes review BOQ, site survey, and power calculations.
- Design Low-level design for the NFVI and NFV such as SBC, IMS, and UPCF.
- Work on solutions and implement them for new build and expansion service migrations.
- Oversaw migration of critical VoLTE service to core customer network platforms.
- Collaborated with cross-functional teams to implement ASBC, UPCF, and IMS network to the customer EPC and CS network.
- Installing and Configuring, network devices such as routers, switches, servers, storage, and Virtual Network Functions infrastructure such as MDAF(AI Network analyzer), FusionSphere(OpenStack), FusionStage(k8s), eSight, LCM, and OSMU.
- Installing and commissioning Virtual Network Functions such as NewCalling(AI Implemented Core Network), SBC, CSCF, ATS, and vDRA.
- Managed virtualized server environment spanning multiple data centers.

Education

Telkom University **2022**
Bachelor of Engineering in Telecommunications
GPA: 3.52 (Cumlaude)

Certifications

- Huawei - Competency and Qualification L2 Service Engineer for Core Network product
- Huawei - 2024 Overseas VoLTE and SVC delivery Training (Project delivery scenario)
- Huawei - Project Management General
- Kotlin - Mobile Development
- Tensorflow - DeepLearning.AI with TensorFlow

Projects

Huawei - Core IMS Project **2025**

XL MDAF

- Lead MDAF proof of concept project, this product have the ability to analyze the customer whole network and make simulations from the analyze result.

XL New Calling

- Lead New Calling proof of concept project, this product have the ability to live-translate during video call, and have a AI avatar that response to human interactions

XL IMS Cibitung 2

- Newly build IMS Cibitung 02, with the purpose of handling VoLTE traffic and MO Anchoring traffic.
- Responsible for working on the BOQ review, site planning, LLD design, L2 LAN switch, L3 router, NFVI commissioning, software commissioning, surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

XL MGCF Mo Anchoring

- Newly build MGCF for traffic Mo Anchoring, working together with CBTIMS02 to serve the MO Anchoring traffic.
- Responsible for working on the BOQ review, site planning, LLD design, L2 LAN switch, L3 router, NFVI commissioning, software commissioning, surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

XL USC new build

- Newly build USC Cibitung and Surabaya, to swap the old vDRA VoLTE Cibitung and Surabaya.
- Lead in the project, responsible for working on the BOQ review, site planning, LLD design, L2 LAN switch, L3 router, NFVI commissioning, software commissioning, surrounding interworking, service ATP, software ATP, and service migrations.

XL IMS Pekanbaru

- To serve Sumatra's north and south MME area, the APN IMS existing traffic will be migrated from IMS Cibitung to IMS Pekanbaru.
- Responsible for working on the L2 LAN switch, L3 router, NFVI commissioning, software commissioning, surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

XL UPCF VoLTE Expansion

- To compensate for the growing VoLTE service, increase the PCRF capacity.
- Responsible for BOQ review, site planning, LLD design, working on L2 LAN switch, NFVI commissioning, Software commissioning, Surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

Telkom Indonesia xGCF Modernization

- Support of site survey, activity planning, and Hardware installations.

XL UPCF VoLTE Expansion

- Review BoQ, site survey, order material, and make LLD.
- Lead site Installations and

Huawei - Core IMS Project **2024**

TELKOM SBC New Build

- Serve as SBC for VoBB network, lead installations, and commissioning at TREG-3 Cirebon.
- Responsible for designing LLD design, L2 LAN switch installation, L3 router installation, NFVI commissioning, software commissioning, support surrounding interworking, service ATP, software ATP, and help service migrations.

XL IMS APN enablement on JABO EPC network

- Enablement of IMS APN in multiple CP and UP in the EPC core.
- Responsible for Service testing for the VOLTE service that has correlations with EPC core such as IMS APN, VoLTE voice call session, and VoLTE video call session.

XL ENS Expansion

- To compensate for the increasing number of VoLTE subscribers the ENUM server is expanded.
- Responsible for Working at the L2 lanswitch, L3 Router, NFVI commisioning.

XL SPS Expansion

- To compensate for the increasing number of VoLTE subscribers the vDRA server is expanded.
- Responsible for working at the NFVI and supporting migrations.

XL IMS Pekanbaru

- To serve Sumatra's north and south MME area, the APN IMS existing traffic will be migrated from IMS Cibitung to IMS Pekanbaru.
- Responsible for Working at L2 LAN switch, L3 Router, NFVI commissioning, Software commissioning, Surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

XL UPCF VoLTE Expansion

- To compensate for the growing VoLTE service, increase the PCRF capacity.
- Responsible for BOQ review, site planning, LLD design, Working at L2 LAN switch, NFVI commissioning, Software commissioning, Surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

XL UPCF VoLTE Expansion

- Review BoQ, site survey, order material, and make LLD.

Huawei - Core XL IMS Project **2023**

XL SBC versions upgrade

- Upgrade 4 SBC to compensate for the new feature.
- Planning upgrade, making upgrade solutions, executing upgrade solutions.

XL SPS Expansion

- To compensate for the increasing number of VoLTE subscribers the vDRA server is expanded, mainly to expand the link to the UPCF server.
- working at the NFVI and support migrations.

XL UPCF VOLTE New Build

- New build for the UPCF for VoLTE service, mainly to expand the capacity for VoLTE service
- working at the NFVI and support migrations.

XL IMS Denpasar New Build

- Mainly interworking with Ericsson device, Ericsson MGCF, and Ericsson PGW.
- To compensate XL subscribers in the east area so it will Load share to the existing IMS.
- Responsible for Working at L2 LAN switch, L3 Router, NFVI commissioning, Software commissioning, Surrounding interworking, service ATP, software ATP, CDR ATP, and service migrations.

Huawei - Core XL IMS Project **2022**

XL SBC Expansion

- With the traffic increase for VoLTE service, there is a need to increase SBC VM to compensate for IMS service traffic.
- working at service testing and support traffic migrations.

XL RO interface implementations

- Customer wants to implement OCS to the existing IMS.
- Working at developing tools to meet customer deadlines.